

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO2 – Manipulation	Assessment:	Assessment Tool 1 – ANALYTICAL PROBLEM SOLVING
Weightage:	50% of CIA		
CO5 Statement:	Design inflectional, derivational, finite-state parsing, stemming algorithms and for analyse morphological methods. (BL-3)		
Student Name:	Harshavardhan R V	Reg. No.:	192472163
Section / Batch:	CSE-AI	Date:	1-8-2026

Code of Conduct

I HARSHAVARDHAN RV(192472163) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Harshavardhan RV

Instructions

- Read all questions carefully before attempting the answers.
- Answer all five questions. Each question carries equal marks.
- Show all intermediate steps, calculations, probability computations, tagging decisions, and justifications wherever applicable.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
English Word Classes, Part of Speech Tagging	Morphological decomposition of connected, connecting, and connection; identification of roots, suffixes, inflectional vs. derivational forms, and normalization for document indexing.	Q1
English Word Classes, Tagsets for English, Rule-Based Part of Speech Tagging	Morphological parsing of unhappy, happiness, and happily; identification of prefixes, suffixes, base forms, and semantic effects of derivational morphology.	Q2
Counting Words, Part of Speech Tagging, English Word Classes	Stemming and root extraction for played, player, and playing; handling grammatical variations and word formation changes for search preprocessing.	Q3
Finite-State Morphological Analysis, Rule-Based Part of Speech Tagging, Transformation-Based Tagging	Finite-state parsing of writes, writing, and written; modeling regular and irregular verb inflections and root normalization.	Q4
English Word Classes, Rule-Based Part of Speech Tagging, Counting Words	Porter Stemmer-based processing of relational, relation, and relate; application of derivational suffix removal and stem extraction for retrieval.	Q5

Annexure A – ANALYTICAL PROBLEM SOLVING

1. Given the input words: “connected”, “connecting”, “connection”, design a morphological analysis pipeline for a document indexing system.
Apply rules to decompose each word into root and suffix components.
Differentiate between inflectional and derivational forms within the dataset.
Normalize all variants to a common base form for efficient retrieval.
Determine the parsed structure and normalized output for each word.

1. Morphological Analysis Pipeline

Input Words: connected, connecting, connection

Step 1: Rule-Based Decomposition

Input Word	Root	Suffix	Type
connected	connect	-ed	Inflectional
connecting	connect	-ing	Inflectional
connection	connect	-ion	Derivational

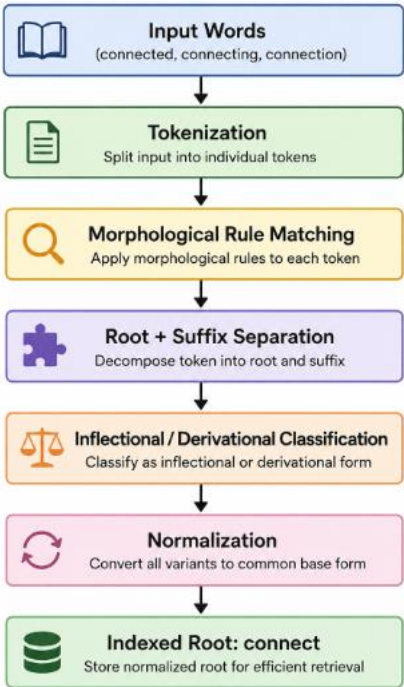
Step 2: Classification

- connected → Past tense (Inflectional)
- connecting → Present participle (Inflectional)
- connection → Noun formation (Derivational)

Step 3: Normalization

Input	Normalized Base Form
connected	connect
connecting	connect
connection	connect

Morphological Analysis Pipeline



Final Parsed Output

Word	Parsed Structure	Normalized Output
connected	connect + ed	connect
connecting	connect + ing	connect
connection	connect + ion	connect

2.Given the input words: “unhappy”, “happiness”, “happily”, design a morphological parsing module for sentiment analysis.
Identify prefixes, suffixes, and base forms using rule-based decomposition.
Ensure semantic consistency across derived word forms.
Classify each transformation as inflectional or derivational.
Produce the root and morphological breakdown for each word.

2. Morphological Parsing Module

Input Words: unhappy, happiness, happily

Step 1: Rule-Based Decomposition

Input Word	Prefix	Root	Suffix
unhappy	un	happy	—
happiness	—	happy	-ness
happily	—	happy	-ly

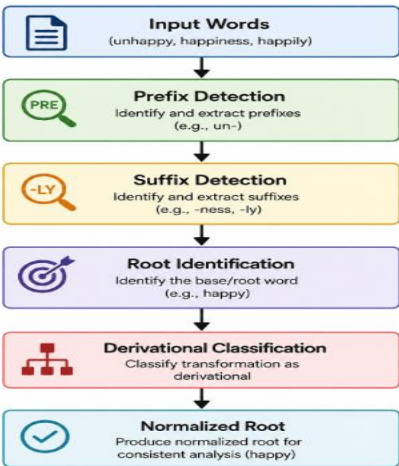
Step 2: Classification

Word	Transformation Type
unhappy	Derivational
happiness	Derivational
happily	Derivational

Step 3: Semantic Consistency

All words share the same base meaning "happy", enabling consistent sentiment analysis.

Morphological Parsing Flow



Final Output

Word	Morphological Breakdown	Root
unhappy	un + happy	happy
happiness	happy + ness	happy
happily	happy + ly	happy

3. Given the input words: “played”, “player”, “playing”, design a stemming-based preprocessing module for a search engine.
Apply morphological rules to strip affixes and identify the base form.
Handle both grammatical variations and word formation changes.
Ensure consistent root extraction across all forms.
Determine the stem and classification for each input word.

3. Stemming-Based Preprocessing Module

Input Words: played, player, playing

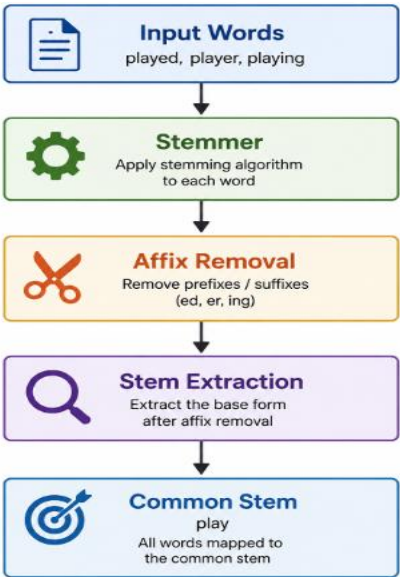
Step 1: Apply Stemming Rules

Input Word	Removed Affix	Stem
played	-ed	play
player	-er	play
playing	-ing	play

Step 2: Classification

Word	Type
played	Inflectional
player	Derivational
playing	Inflectional

Stemming Flow



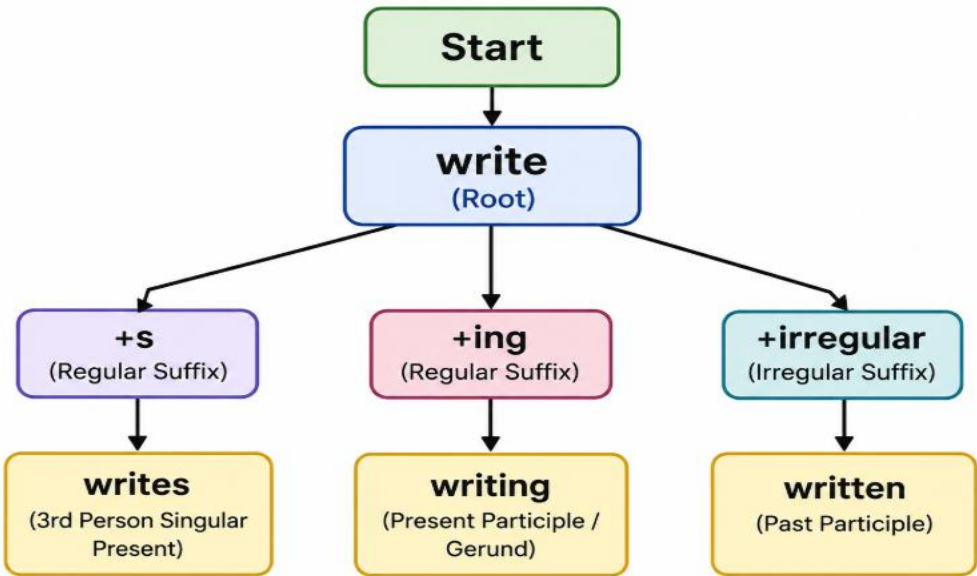
Final Output

Word	Stem	Classification
played	play	Inflectional
player	play	Derivational
playing	play	Inflectional

4. Given the input words: “writes”, “writing”, “written”, design a finite-state morphological parsing module for a text processing system.
Apply state transitions to represent suffix transformations and irregular forms in verb inflection.
Differentiate between regular and irregular inflectional patterns within the dataset.
Ensure consistent mapping of all forms to a common root representation.
Determine the state transitions, morphological breakdown, and normalized output for each word.

4. Input Words: writes, writing, written

Finite-State Transitions



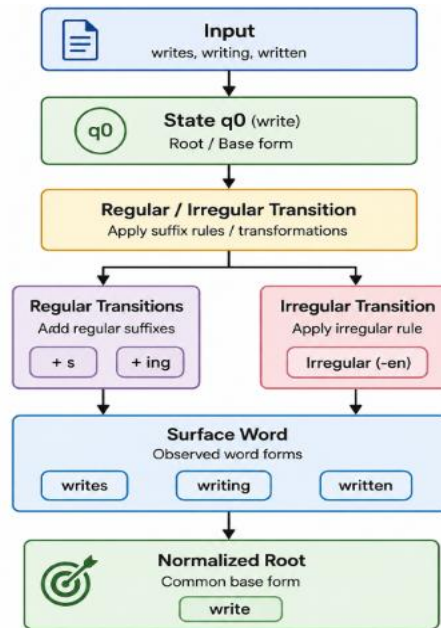
Morphological Breakdown

Input Word	Root	Suffix/Transformation	Pattern
writes	write	-s	Regular Inflection
writing	write	-ing	Regular Inflection
written	write	irregular (-en)	Irregular Inflection

Normalized Output

Word	Normalized Root
writes	write
writing	write
written	Write

Finite-State Parsing Flow



5. Given the input words: “relational”, “relation”, “relate”, design a Porter Stemmer-based preprocessing module for document retrieval.

Apply stemming rules step-by-step to remove derivational suffixes and obtain base forms.

Handle variations in suffix patterns while preserving semantic consistency.

Ensure uniform root extraction across all derived forms.

Determine the stemming steps, intermediate forms, and final stem for each input word.

5. Porter Stemmer-Based Preprocessing

Input Words: relational, relation, relate

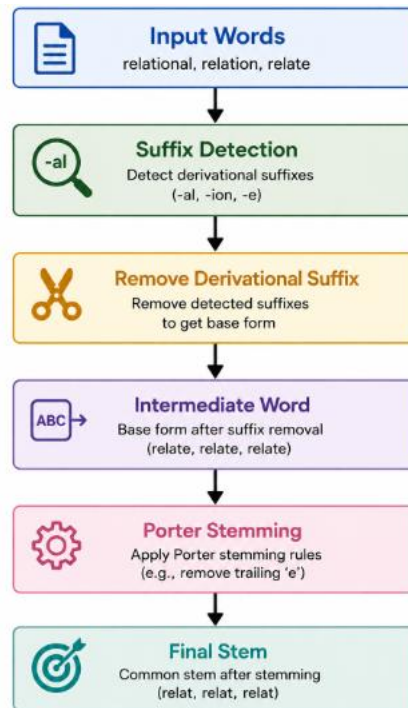
Step-by-Step Stemming

<i>Input Word</i>	<i>Step 1</i>	<i>Step 2</i>	<i>Final Stem</i>
<i>relational</i>	relational → relate	relate → relat	relat
<i>relation</i>	relation → relate	relate → relat	relat
<i>relate</i>	relate	relate → relat	relat

Intermediate Forms

<i>Word</i>	<i>Intermediate Form</i>	<i>Final Stem</i>
<i>relational</i>	relate	relat
<i>relation</i>	relate	relat
<i>relate</i>	relate	relat

Porter Stemmer Flow



Final Output

<i>Word</i>	<i>Stemming Steps</i>	<i>Final Stem</i>
<i>relational</i>	relational → relate → relat	relat
<i>relation</i>	relation → relate → relat	relat
<i>relate</i>	relate → relat	relat

Rubrics for Calculation

Numerical Problems					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Q. No. / Criteria Level	Problem Understanding	Method Selection	Solution Process	Accuracy of Calculation	Interpretation of Results	Sub. Total	Total %
1							
2							
3							
4							
5							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____